

The objective of this project is to analyze superstore sales using EDA and find out the factors affecting sales

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

1. LOADING DATA

```
df = pd.read_csv("superstore_final_dataset (1).csv", encoding='latin1')
```

```
df.head()
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9800 entries, 0 to 9799
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Row_ID                9800 non-null  int64
1   Order_ID              9800 non-null  object
2   Order_Date            9800 non-null  object
3   Ship_Date             9800 non-null  object
4   Ship_Mode             9800 non-null  object
5   Customer_ID           9800 non-null  object
6   Customer_Name         9800 non-null  object
7   Segment               9800 non-null  object
8   Country               9800 non-null  object
9   City                  9800 non-null  object
10  State                 9800 non-null  object
11  Postal_Code           9789 non-null  float64
12  Region                9800 non-null  object
13  Product_ID            9800 non-null  object
14  Category              9800 non-null  object
15  Sub_Category          9800 non-null  object
16  Product_Name          9800 non-null  object
17  Sales                 9800 non-null  float64
dtypes: float64(2), int64(1), object(15)
memory usage: 1.3+ MB
```

	Row_ID	Postal_Code	Sales
count	9800.000000	9789.000000	9800.000000
mean	4900.500000	55273.322403	230.769059
std	2829.160653	32041.223413	626.651875
min	1.000000	1040.000000	0.444000
25%	2450.750000	23223.000000	17.248000
50%	4900.500000	58103.000000	54.490000
75%	7350.250000	90008.000000	210.605000
max	9800.000000	99301.000000	22638.480000

```
df.isnull().sum()
```

	0
Row_ID	0
Order_ID	0
Order_Date	0
Ship_Date	0
Ship_Mode	0
Customer_ID	0
Customer_Name	0
Segment	0
Country	0
City	0
State	0
Postal_Code	11
Region	0
Product_ID	0
Category	0
Sub_Category	0
Product_Name	0
Sales	0

dtype: int64

```
df.dropna(inplace=True)
```

```
df.drop_duplicates(inplace=True)
```

NO DUPLICATE DATA

```
df['Order_Date'] = pd.to_datetime(df['Order_Date'], dayfirst=True)
```

```
df['Year'] = df['Order_Date'].dt.year
```

```
df['Month'] = df['Order_Date'].dt.month
```

```
df['Month_Name'] = df['Order_Date'].dt.month_name()
```

2. CREATING THE KPI's

```
total_revenue = df['Sales'].sum()
```

```
print(total_revenue)
```

```
2252607.4127
```

```
total_orders = df['Order_ID'].nunique()
```

```
print("Total Orders:", total_orders)
```

```
Total Orders: 4916
```

```
total_customers = df['Customer_ID'].nunique()

print("Total Customers:", total_customers)
```

```
Total Customers: 793
```

```
aov = total_revenue / total_orders

print("Average Order Value:", round(aov, 2))
```

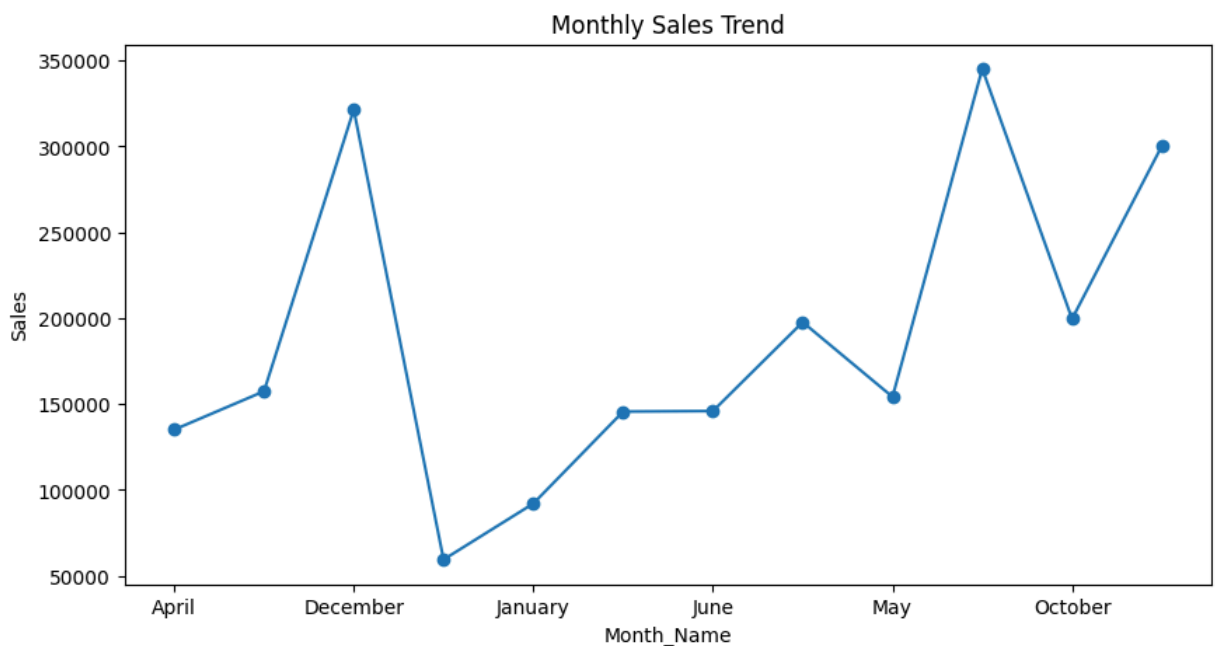
```
Average Order Value: 458.22
```

3. monthly sales

```
monthly_sales = df.groupby('Month_Name')['Sales'].sum()

monthly_sales.plot(
    kind='line',
    marker='o',
    figsize=(10,5)
)

plt.title("Monthly Sales Trend")
plt.ylabel("Sales")
plt.show()
```



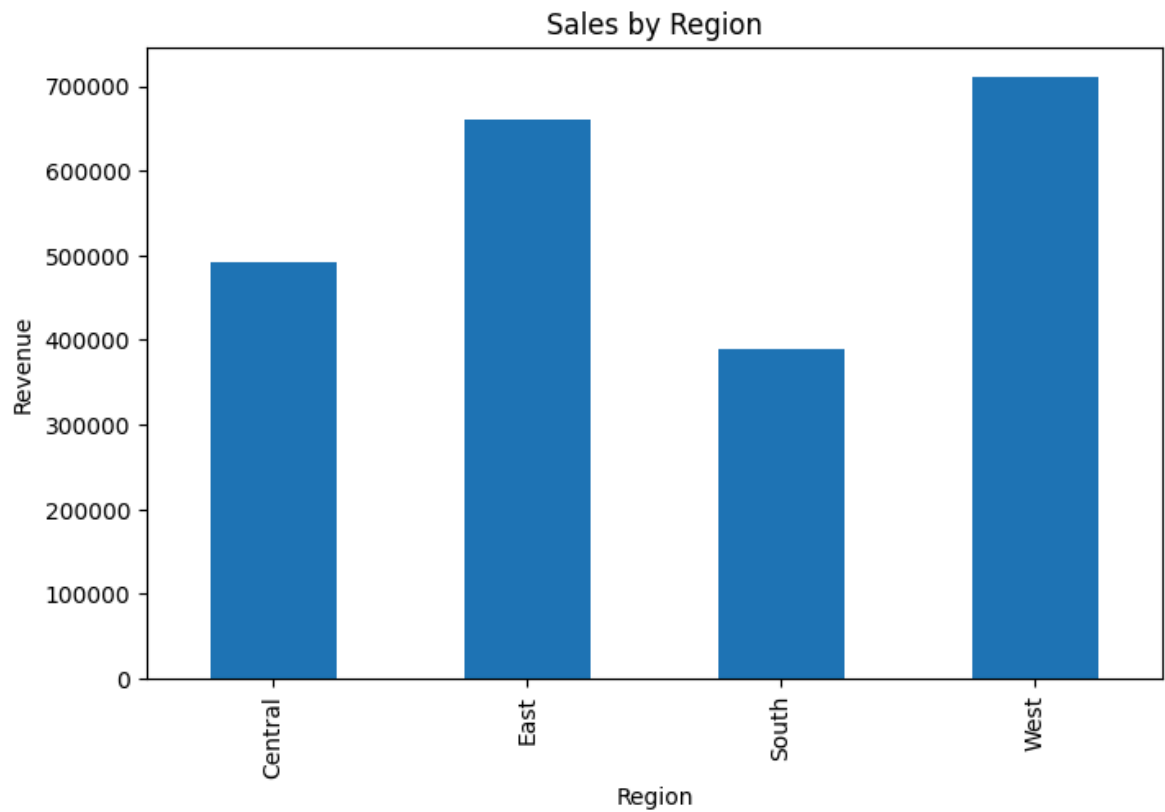
4. regional analysis

```
region_sales = df.groupby('Region')['Sales'].sum()

region_sales.plot(
    kind='bar',
    figsize=(8,5)
)

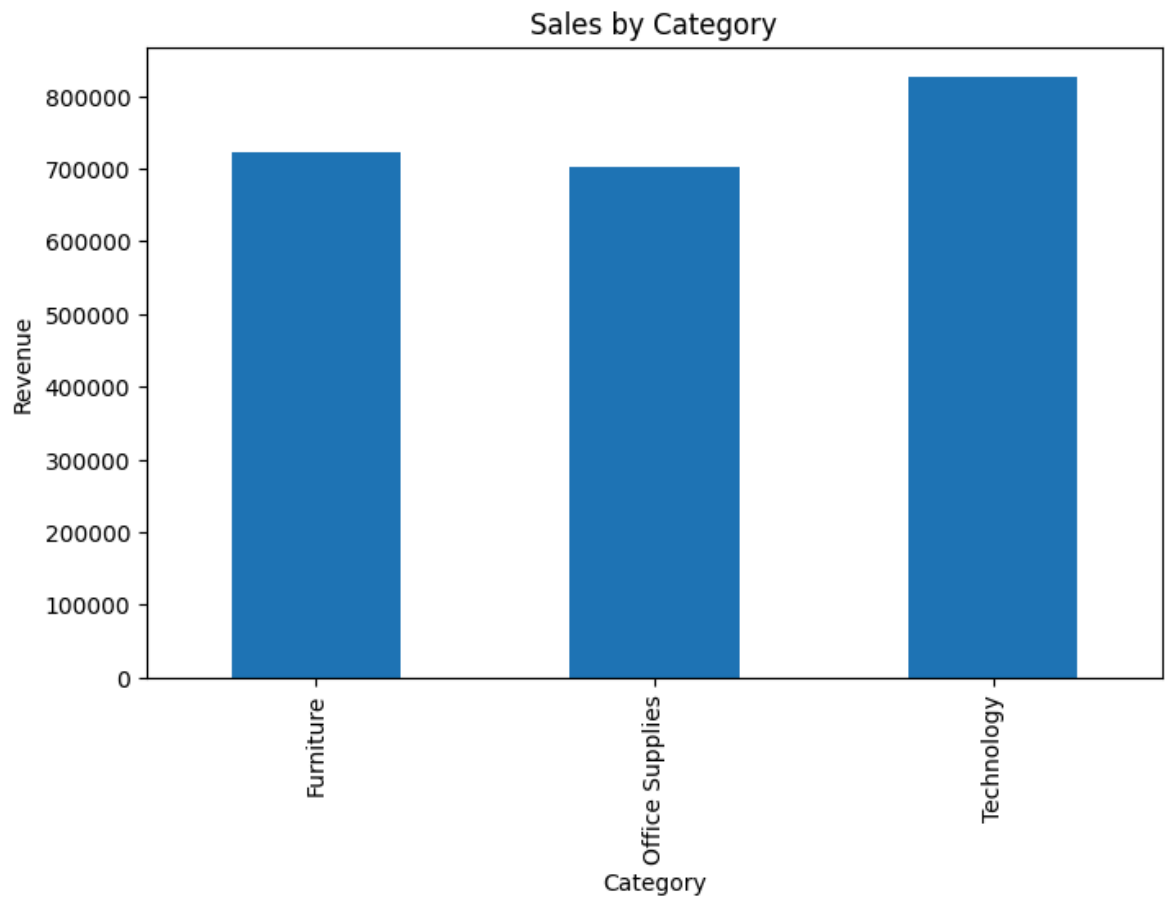
plt.title("Sales by Region")
```

```
plt.ylabel("Revenue")  
plt.show()
```

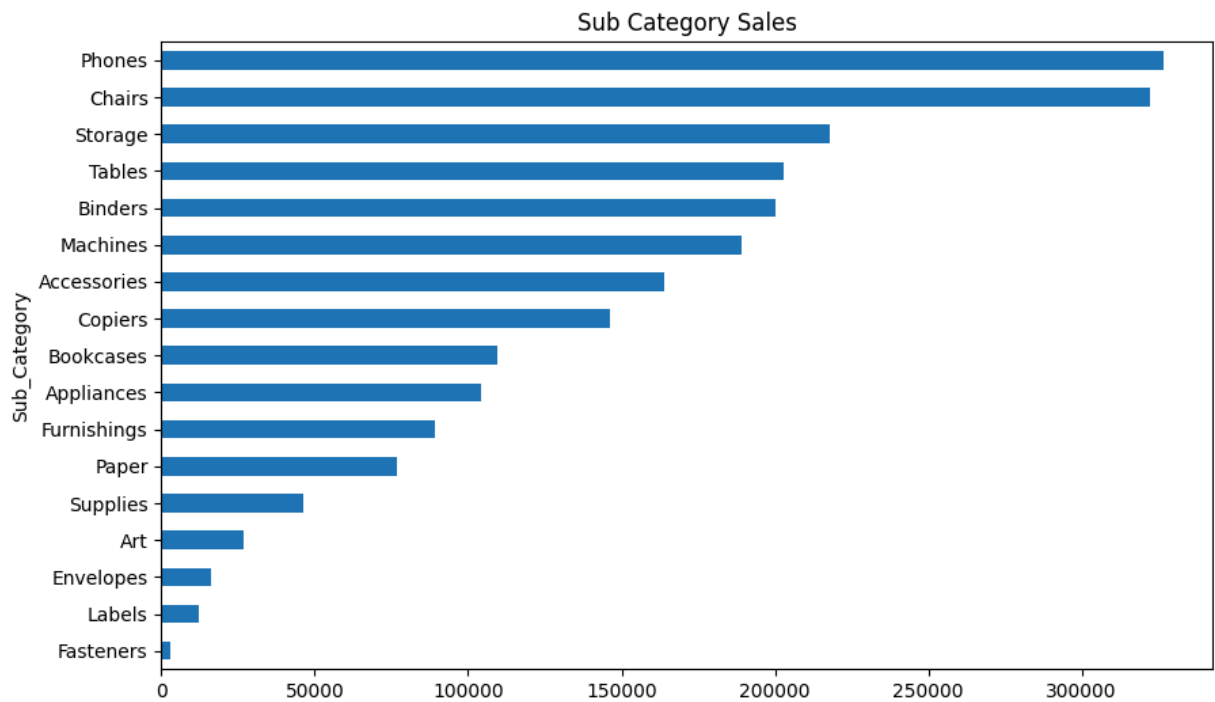


5. catogory analysis

```
category_sales = df.groupby('Category')['Sales'].sum()  
  
category_sales.plot(  
    kind='bar',  
    figsize=(8,5)  
)  
  
plt.title("Sales by Category")  
plt.ylabel("Revenue")  
plt.show()
```



```
subcategory_sales = df.groupby(  
    'Sub_Category'  
)['Sales'].sum()  
  
subcategory_sales.sort_values().plot(  
    kind='barh',  
    figsize=(10,6)  
)  
  
plt.title("Sub Category Sales")  
plt.show()
```



6. top 10 products and its visualization

```
top_products = df.groupby(
    'Product_Name'
)['Sales'].sum()

top10 = top_products.sort_values(
    ascending=False
).head(10)

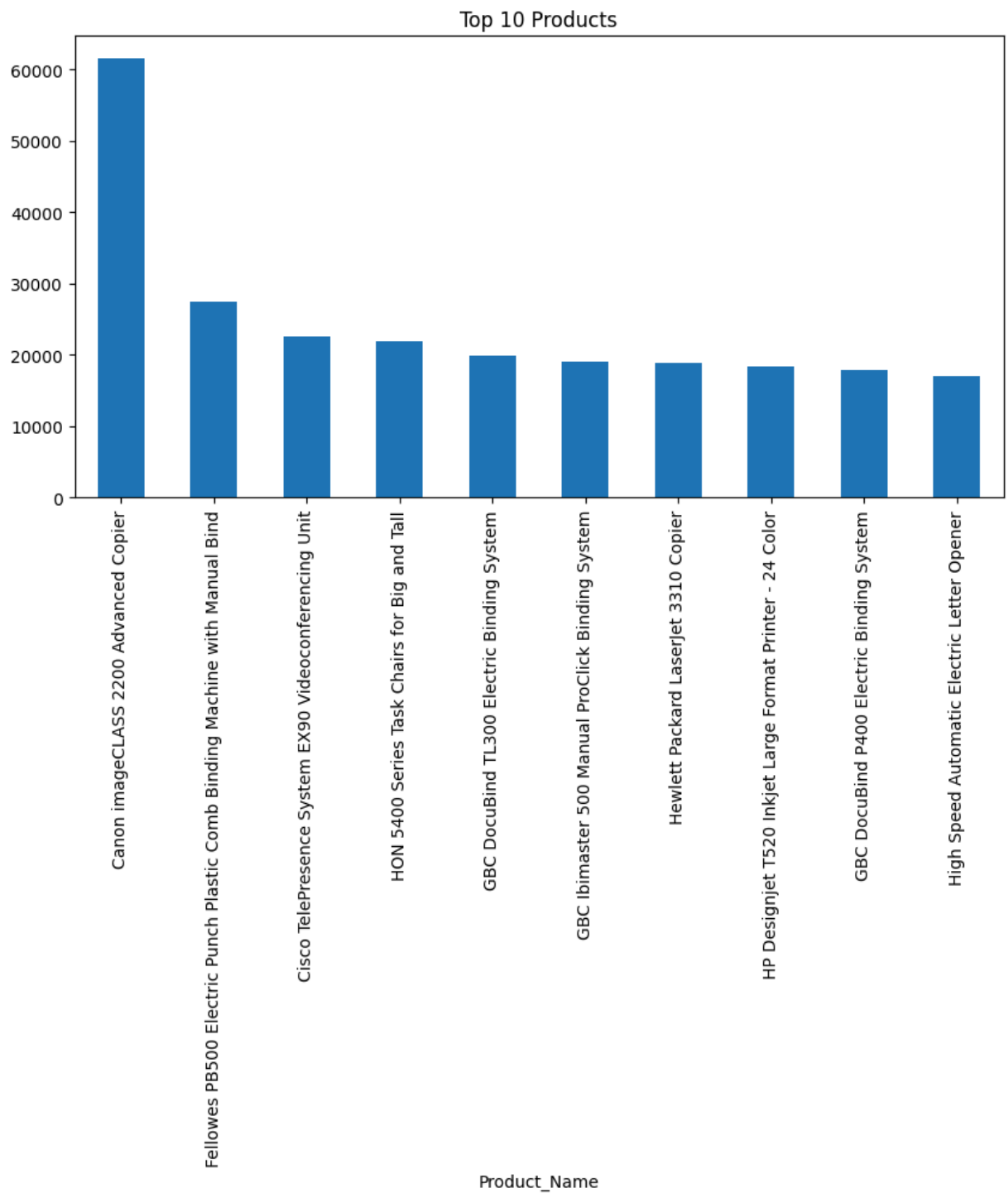
print(top10)
```

Product_Name	
Canon imageCLASS 2200 Advanced Copier	61599.824
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	27453.384
Cisco TelePresence System EX90 Videoconferencing Unit	22638.480
HON 5400 Series Task Chairs for Big and Tall	21870.576
GBC DocuBind TL300 Electric Binding System	19823.479
GBC Ibimaster 500 Manual ProClick Binding System	19024.500
Hewlett Packard LaserJet 3310 Copier	18839.686
HP Designjet T520 Inkjet Large Format Printer - 24 Color	18374.895
GBC DocuBind P400 Electric Binding System	17965.068
High Speed Automatic Electric Letter Opener	17030.312

Name: Sales, dtype: float64

```
top10.plot(
    kind='bar',
    figsize=(10,5)
)

plt.title("Top 10 Products")
plt.show()
```



```
bottom10 = df.groupby(
    'Product_Name'
)['Sales'].sum().sort_values().head(10)

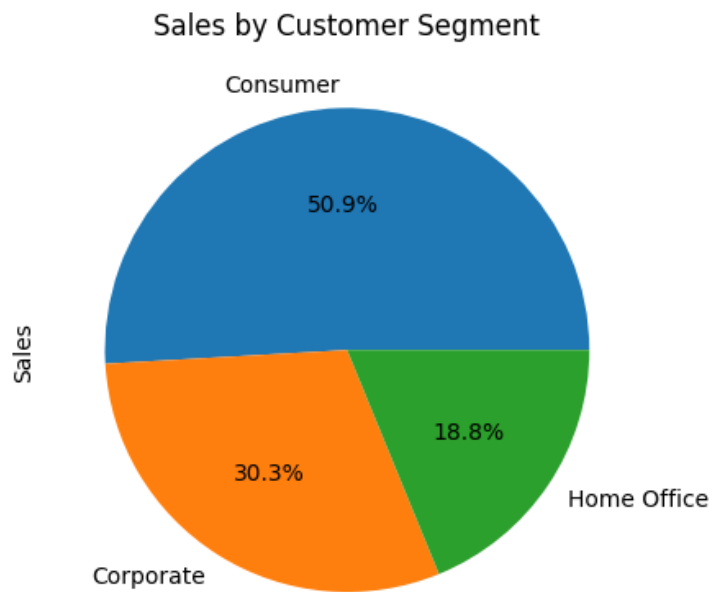
print(bottom10)
```

Product_Name	
Eureka Disposable Bags for Sanitaire Vibra Groomer I Upright Vac	1.624
Avery 5	5.760
Xerox 20	6.480
Grip Seal Envelopes	7.072
Acme Serrated Blade Letter Opener	7.632
Avery Hi-Liter Pen Style Six-Color Fluorescent Set	7.700
Avery Hi-Liter Comfort Grip Fluorescent Highlighter, Yellow Ink	7.800
Xerox 1989	7.968
Stockwell Gold Paper Clips	8.096

Acco Economy Flexible Poly Round Ring Binder
Name: Sales, dtype: float64

8.352

```
segment_sales = df.groupby(  
    'Segment'  
)['Sales'].sum()  
  
segment_sales.plot(  
    kind='pie',  
    autopct='%1.1f%%'  
)  
  
plt.title("Sales by Customer Segment")  
plt.show()
```



```
state_sales = df.groupby(  
    'State'  
)['Sales'].sum().sort_values(  
    ascending=False  
)  
.head(10)  
  
state_sales.plot(  
    kind='bar'  
)  
  
plt.title("Top 10 States by Sales")  
plt.show()
```